Before the Form

Why a Traversal-Based Semantic System Is Close to, but Not Identical with, Spencer-Brown's Laws of Form

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A conceptual comparison written from the standpoint of the author's semantic traversal system. The paper distinguishes a completed form of relation from relation as an operation in passage.

Abstract

G. Spencer-Brown's Laws of Form begins from one of the deepest possible starting points: the instruction to draw a distinction. It does not begin with a conventional object, proposition, particle, or number. In this respect, it comes extremely close to the foundation developed in my semantic word system. However, the two systems do not make the same foundational move. Spencer-Brown begins with an operation, but rapidly contracts that operation into a mark that can simultaneously function as a boundary, a state, an indication, a value, an instruction, and a formal token. The relation established by the distinction is then treated primarily as a relation of continence: something is inside, outside, marked, unmarked, contained, or not contained. My system does not begin by installing even this minimal form. It treats relation as the active traversal through which distinction is carried, transformed, recovered, and closed. No object is placed at the beginning of the system. What we later call an object is the invariant appearance of a traversal that has successfully preserved enough of itself to close. The difference is therefore not merely between two notations. It is the difference between relation as formed structure and relating as the operation from which structure becomes possible.

Contents

1. The Nearness of the Two Systems

Spencer-Brown begins with the command to draw a distinction. This is remarkably close to my own starting point because it rejects the usual assumption that reality must first be composed of independently existing objects. Rather than beginning with two things and then adding a relation between them, the system begins with an act.

$$A,\ B,\ A\,R\,B \quad \rightarrow \quad \text{distinguish}$$

The distinction comes before the ordinary logical entities that may later occupy its sides. In this respect, Laws of Form reaches beneath ordinary propositional logic. It asks what must occur before there can be a proposition, value, or indicated thing.

That is why the system feels so close to mine. Both systems recognise that description cannot begin with a fully established object. Something must first become distinguishable. There must be some operation through which a difference can appear. But the exact treatment of that appearance is where the systems separate.

2. Spencer-Brown Does Not Begin with an Ordinary Object

It would be inaccurate to say that Spencer-Brown simply begins with an object. He does not begin with a chair, a particle, a set, a number, or even a truth value. He begins with an injunction. The opening movement is operational rather than representational.

The important criticism is therefore subtler: Spencer-Brown begins without an ordinary object, but he then contracts the distinguishing act into a form capable of being treated as an object-like formal unit.

- the marked state
- the unmarked state
- the mark
- the token of the mark
- the form
- the expression
- the value of the expression

The original operation is compressed into something that can be repeated, nested, crossed, evaluated, and compared. This contraction is the source of the enormous power of Laws of Form. It is also the point at which Spencer-Brown's foundation becomes different from mine.

3. The Mark Freezes Several Stages into One Sign

Spencer-Brown's mark is deliberately overloaded. The same sign can stand for the drawing of a boundary, the boundary drawn, the state indicated, the instruction to cross, and the value reached through crossing.

$$\text{act} \ \equiv\ \text{boundary} \ \equiv\ \text{instruction} \ \equiv\ \text{state} \ \equiv\ \text{value}$$

This is elegant because it places an enormous amount of logical structure into one symbol. But it also means that the system does not keep the stages of the operation distinct. Once this contraction is made, the calculus can operate on forms as if the relevant act has already been successfully completed.

The mark remembers that a distinction occurred, but it does not internally display the entire passage through which the distinction was maintained. It records the form of the distinction. My system is intended to preserve the traversal that makes the form recoverable.

4. Continence and Traversal

The clearest difference appears in the treatment of relation. Spencer-Brown's primary relation is continence. A cross contains what lies on its inside and does not contain what lies outside it. One cross may stand inside another, outside another, beside another, or at some nesting depth.

$$\text{Spencer-Brown:}\ \ \text{relation as arrangement}$$

$$\text{Semantic traversal:}\ \ \text{relating as operation}$$

In Spencer-Brown, a form is contained or not contained. In my system, relation must move, preserve, transmit, transform, resolve, and potentially return. The question is not only: on which side of the distinction does something stand? The more primitive question is: what must happen for distinguishability to survive its passage through relation?

5. No Object Is Installed in the Semantic Word System

My semantic word system does not begin with x and then apply operations to x. Even writing F(x) risks quietly installing an object before the system has earned the right to describe one.

$$\text{Persist} \to \text{Distinguish} \to \text{Relate} \to \text{Propagate} \to \text{Resolve} \to \text{Recurse} \to \text{Close}$$

These words do not initially describe properties of an object. They describe the necessary stages through which anything capable of later appearing as an object must pass. There is no primitive noun underneath the sequence. There is persistence, distinguishing, relating, propagation, resolution, recursion, and closure.

> ***Objecthood is not the material supplied to the operation. It is the appearance produced when the operation has closed with sufficient invariance.***

$$\text{object} = \text{stable appearance of a closed traversal}$$

An object is not what enters the system. It is what the system permits us to name after enough semantic structure has been preserved and recovered.

6. Identity Is Recovered, Not Assumed

In ordinary mathematics, identity is normally installed from the beginning as x = x. The symbol is already assumed to possess enough unity that it can be referred to repeatedly. Spencer-Brown moves earlier than this by grounding indication in distinction, but once the mark is copied as a token its formal identity is supported by the laws of calling and crossing.

My system places identity later. Identity is not what permits traversal to begin. Identity becomes demonstrable when traversal can recover continuity across change.

$$\tau : P \to D \to R \to G \to V \to K \to C$$

Here, τ is not the motion of a previously defined object. It is the completed operational history. Only after closure can two traversals be treated as equivalent when they preserve the same recoverable semantic structure.

$$\tau_1 \sim \tau_2 \quad \text{and} \quad O := [\tau]\sim$$

This does not mean that an object secretly existed before the traversal. It means that several traversals can be recognised as presentations of one invariant pattern. Identity is therefore retrospective: it remained recoverable, and therefore it may be described as an object.

7. Form Is a Closure, Not a Starting Entity

Spencer-Brown makes form arise through distinction, but once form has arisen the calculus operates through arrangements and values of those forms. In my system, form must remain subordinate to traversal. A form is not self-sufficient structure. It is the visible closure of a process.

$$\text{form} = \text{what traversal looks like when its passage is compressed}$$

A static form is a traversal whose internal movement is no longer being displayed. This does not make the form unreal. It makes the form derivative. When persistence, distinction, relation, transmission, resolution, and closure are compressed, the result may look object-like. When they are unfolded, the object disappears into the operational structure that produces it.

8. Re-entry Approaches Traversal, but Does Not Ground the System in It

Spencer-Brown eventually considers re-entry, recursion, oscillation, and the relation between observer and observed form. This is where Laws of Form comes nearest to a traversal-based interpretation. Re-entry reveals that a form cannot always be understood as a static marked or unmarked value; it may require repeated evaluation or temporal unfolding.

But re-entry is introduced after the primary form, its states, and its relations have already been established. Process enters as a consequence of what happens when the form refers back into itself. In my system, traversal does not enter later as the behaviour of an already constituted form. Traversal is primitive.

$$\text{Laws of Form: distinction} \rightarrow \text{form} \rightarrow \text{re-entry}$$

$$\text{Semantic traversal: operation} \rightarrow \text{traversal} \rightarrow \text{closure} \rightarrow \text{form}$$

> ***Spencer-Brown allows form to become process. My system allows process to become form.***

9. This Is Not a Rejection of Laws of Form

The argument is not that Spencer-Brown failed to understand distinction. Laws of Form reaches closer to the operational origin of logic than almost any conventional logical system. It recognises that indication requires distinction and that logical form cannot be separated completely from the act by which a distinction is drawn.

The difference is that Spencer-Brown intentionally contracts that act into a calculable mark. My semantic system refuses to perform that contraction at the foundation. It keeps open the operational distance between drawing, distinguishing, crossing, relating, preserving, recovering, and closing.

$$\text{Laws of Form} = \text{minimal form created by distinction}$$

$$\text{Semantic system} = \text{traversal through which form becomes recoverable}$$

10. Conclusion

Spencer-Brown begins before the conventional object but does not remain entirely before form. The initial act of distinction is contracted into a mark, and the mark becomes a state, boundary, token, instruction, expression, and value. The resulting calculus studies how these forms are arranged, contained, crossed, repeated, and re-entered.

My system begins from a different commitment. No object is installed. No identity is assumed. No relation is treated merely as a completed bridge between two already established sides. The foundation is traversal itself. Distinction must be carried. Relation must propagate. Structure must resolve. Continuity must be recovered. The process must close. Only after closure can something be called a form or object.

> ***Spencer-Brown formalises the distinction once it can be marked. I am trying to formalise what the distinction must do before anything can be marked at all.***

Reference

Spencer-Brown, G. (1969). Laws of Form. London: George Allen & Unwin.